



FAST-FOOD ANALYSIS DASHBOARD

FAST FOOD NUTRITION ANALYSIS
USING POWER BI | SQL | EXCEL

Business Problem

- Fast-food restaurants offer a wide variety of menu items with different nutritional values, making it difficult to compare the nutritional quality of food across multiple restaurant chains.
- Customers increasingly prefer healthier food choices and require clear nutritional information to compare calories, protein, fat, carbohydrates, sugar, fiber, and sodium before purchasing menu items.
- Restaurant management needs meaningful insights into nutritional trends to evaluate existing menu items, identify healthier alternatives, and improve overall menu planning.
- Manually analyzing hundreds of menu items across different restaurants is time-consuming, inefficient, and does not provide actionable business insights.
- An interactive Fast Food Analysis Dashboard is required to compare restaurant performance, identify nutritional patterns, evaluate menu items, and support data-driven decision-making for both customers and restaurant management.

Dataset Overview

Total Menu Items: 510

Number of Restaurants: 8

Data Source: Fast Food Restaurant Database

Key Columns Used:

Restaurant, Item, Calories, Total Fat, Saturated Fat, Trans Fat, Cholesterol, Sodium, Total Carbohydrates, Fiber, Sugar, Protein, Vitamin A, Vitamin C, Calcium.

Purpose:

To analyze the nutritional composition of menu items across different fast-food restaurants, compare restaurant performance, identify healthier food choices, and generate meaningful business insights using Power BI.

Data Cleaning Process

- Connected to the Fast Food database and imported the dataset for analysis.
- Removed unnecessary columns (Satish, age, and salad) as they did not provide analytical value.
- Identified and removed 3 duplicate records to improve data accuracy and eliminate redundant information.
- Handled missing values by replacing the Fiber column with the median value (3) and the Protein column with the median value (25).
- Verified that all nutritional columns were stored in the correct numeric format and prepared the dataset for Power BI visualization and reporting.

FAST FOOD ANALYSIS DASHBOARD

Total Restaurants

8

Total Menu Items

510

Average Calories

532.08

Average Protein (g)

28.00

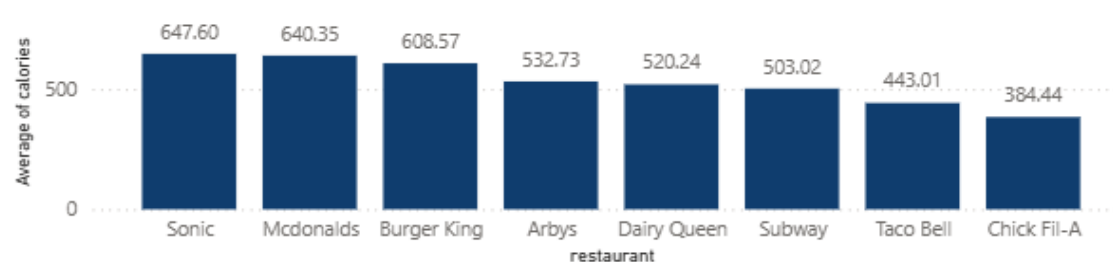
Average Sugar (g)

7.27

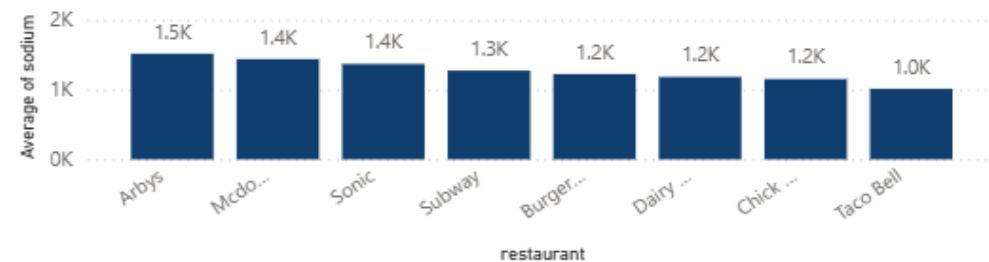
Average Sodium (mg)

1.25K

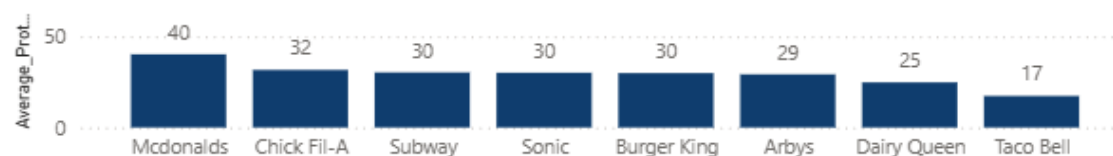
Average Calories by Restaurant



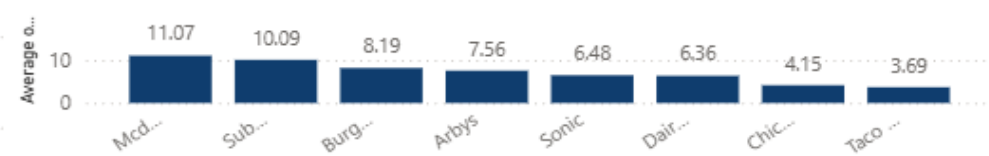
Average Sodium by Restaurant



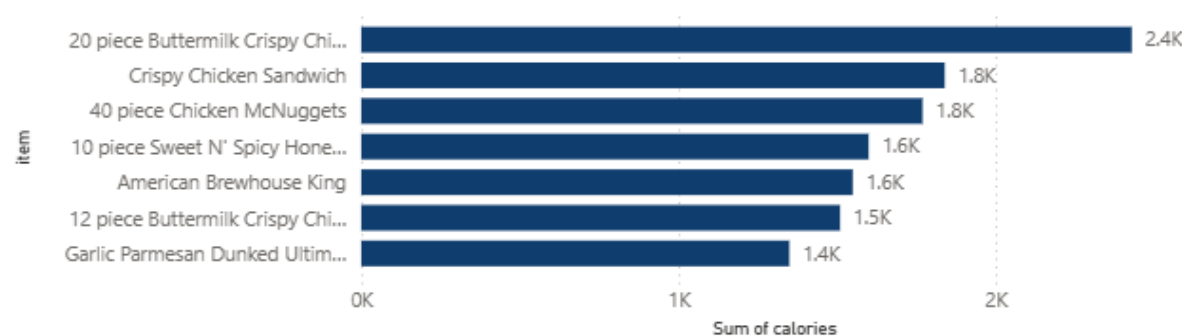
Average Protein by restaurant



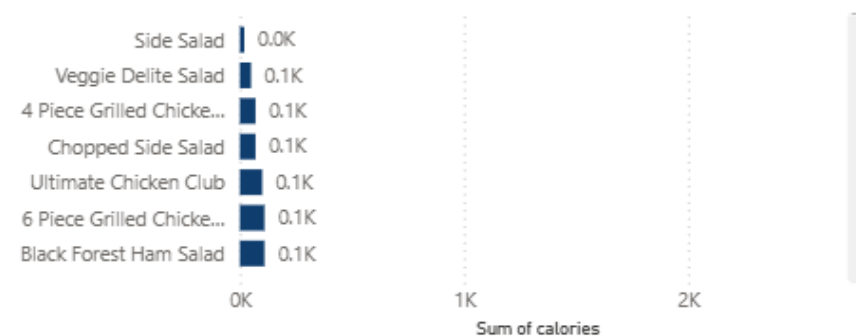
Average sugar by restaurant



Top 10 Highest Calorie Menu Items

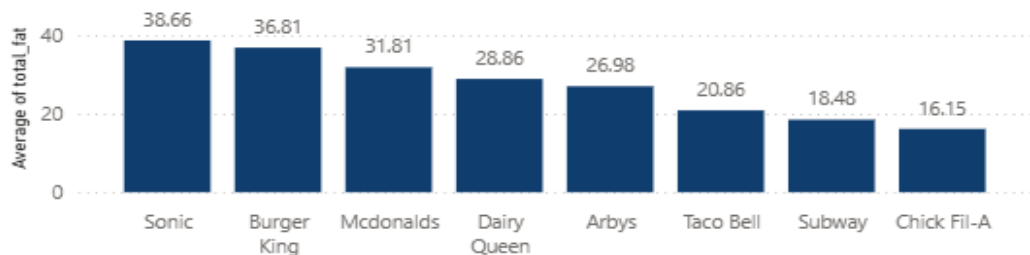


Top 10 Lowest Calorie Menu Items

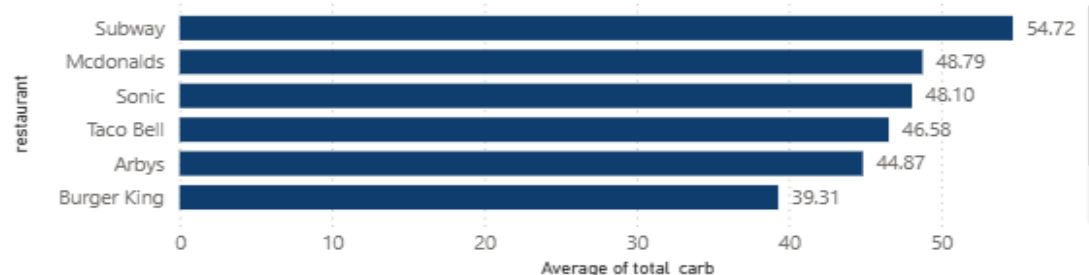


Fast Food Detailed Analysis

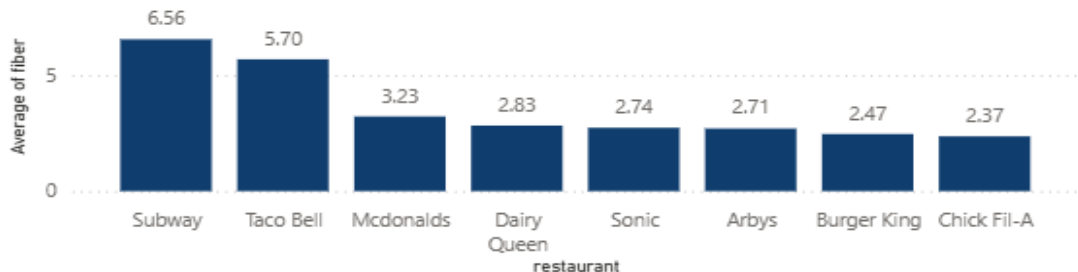
Average Total Fat by Restaurant



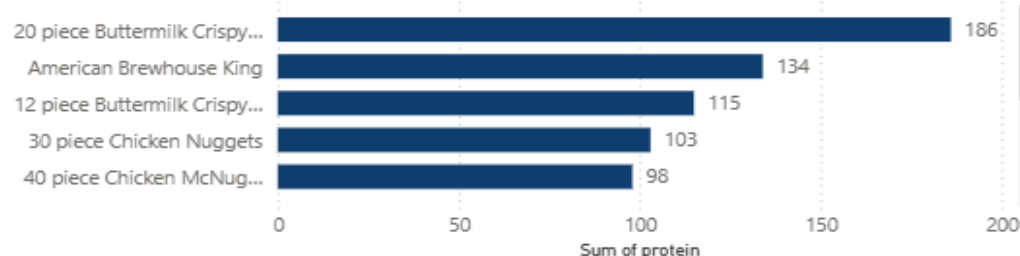
Average Total Carbohydrates by Restaurant



Average Fiber by Restaurant



Top 10 Highest Protein Menu Items



Restaurant Nutrition Comparison

restaurant	Average of fiber	Average of sodium	Average of sugar	Average of calories	Average of protein
Arbys	2.71	1515.27	7.56	532.73	29.25
Burger King	2.47	1223.57	8.19	608.57	29.94
Chick Fil-A	2.37	1151.48	4.15	384.44	31.70
Dairy Queen	2.83	1181.79	6.36	520.24	24.83
Mcdonalds	3.23	1437.89	11.07	640.35	40.30
Sonic	2.74	1368.00	6.48	647.60	30.14
Subway	6.56	1272.97	10.09	503.02	30.31
Total	4.12	1248.39	7.27	532.08	28.00

Menu Item

- ☐ 1 Piece Chick-n-Strips
- ☐ 1/2 lb. FlameThrower® GrillBurger
- ☐ 1/2 lb. GrillBurger with Cheese
- ☐ 1/2 lb.* Cheesy Potato Burrito
- ☐ 1/2 lb.* Combo Burrito
- ☐ 1/4 lb. Bacon Cheese GrillBurger
- ☐ 1/4 lb. GrillBurger with Cheese

Restaurant

- ☐ Arbys
- ☐ Burger King
- ☐ Chick Fil-A
- ☐ Dairy Queen
- ☐ Mcdonalds
- ☐ Sonic
- ☐ Subway

Fast Food Analysis Dashboard Page 1

Dashboard Highlights:

- The dashboard provides a comprehensive overview of the nutritional composition of menu items across multiple fast-food restaurant chains.
- Key Performance Indicators (KPIs) such as Total Restaurants, Total Menu Items, Average Calories, Average Protein, Average Sugar, and Average Sodium provide a quick summary of the dataset.
- Restaurant-wise comparisons of calories, protein, sugar, sodium, fat, carbohydrates, and fiber help identify nutritional differences between restaurant chains.
- Top 10 Highest Calorie and Top 10 Highest Protein menu items were identified to highlight items with extreme nutritional values.
- Interactive slicers allow users to filter the dashboard by restaurant and menu item, enabling dynamic analysis and better decision-making.

Fast Food Analysis Dashboard Page 2

Dashboard Highlights:

- Provides an overall summary of nutritional information across multiple fast-food restaurants.
- Displays key performance indicators including Total Restaurants, Total Menu Items, Average Calories, Average Protein, Average Sugar, and Average Sodium.
- Compares restaurants based on average calories, protein, sugar, and sodium.
- Highlights the Top 10 Highest Calorie and Top 10 Lowest Calorie menu items.
- Interactive slicers allow users to filter the dashboard by restaurant and menu item.

Key Business Insights

- The analysis identified significant differences in calorie, fat, protein, sugar, and sodium content across the eight fast-food restaurant chains.
- Some restaurants consistently offer higher-calorie menu items, while others provide comparatively lower-calorie options, giving customers a wider range of dietary choices.
- High-protein menu items were identified, making them suitable options for customers seeking protein-rich meals and balanced nutrition.
- The dashboard highlights menu items with the highest and lowest calorie values, enabling users to quickly identify energy-dense and lighter food options.
- Interactive dashboard filters allow users to compare restaurants and individual menu items, making nutritional analysis faster, more flexible, and easier to understand.
- The insights generated from this analysis can support healthier menu planning, improve customer awareness, and assist restaurant management in making data-driven decisions.

Recommendations

- Introduce additional low-calorie and balanced meal options to meet the growing demand for healthier food choices.
- Promote high-protein and high-Fiber menu items to attract health-conscious and fitness-focused customers.
- Review menu items with high sodium and fat content and explore opportunities to improve their nutritional profile.
- Display clear nutritional information on menus and digital platforms to help customers make informed purchasing decisions.
- Use dashboard insights regularly to monitor nutritional trends, evaluate menu performance, and support future product development.

Conclusion

- The Fast Food Analysis Dashboard successfully analyzed the nutritional information of menu items across multiple restaurant chains.
- The dashboard provided valuable insights into calories, protein, fat, carbohydrates, sugar, sodium, and Fiber through interactive visualizations.
- The analysis enabled meaningful comparisons between restaurants and helped identify both healthier food choices and high-calorie menu items.
- These insights can support better menu planning, improve customer awareness, and help restaurant management make informed, data-driven decisions.

THANK YOU..!

Presented By
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